Absolutely. Below are **each OS topic separately in proper 5-mark exam format**, using **simple language** so you can learn and write them in the exam.

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1. TYPES OF OPERATING SYSTEMS — 5 MARKS

Definition

An **Operating System (OS)** is system software that manages computer hardware and provides services to application programs.

Types of Operating Systems

1. Batch Operating System
Jobs are collected and executed in groups or batches. There is little or no interaction between the user and the computer during execution.

2. Multiprogramming Operating System
Multiple programs are kept in main memory at the same time. The CPU switches between them to improve CPU utilization.

3. Multitasking Operating System
It allows multiple tasks to run apparently at the same time.
Example: Windows, Linux.

4. Multiprocessing Operating System
It uses two or more processors/CPUs to execute processes simultaneously.

5. Real-Time Operating System
It provides a response within a specified time limit.
Example: Airbag control system.

Conclusion

Different types of operating systems are designed to satisfy different requirements such as **speed, resource utilization, multitasking, and real-time response**.

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2. OS STRUCTURES — 5 MARKS

Definition

OS structure describes how the components and services of an operating system are organized.

Types of OS Structures

1. Monolithic Structure
All major OS services run inside one large kernel.

2. Layered Structure
The OS is divided into different layers. Each layer performs a specific function.

3. Microkernel Structure
Only essential services are kept in the kernel. Other services run outside the kernel.

4. Modular Structure
The OS is divided into separate modules. Modules can be added or removed when required.

5. Hybrid Structure
It combines features of different OS structures to provide better performance and flexibility.

Simple Diagram

Application Programs

↓

System Services

↓

Operating System Kernel

↓

Hardware

Conclusion

OS structures organize the operating system and make it easier to **design, maintain, secure, and improve**.

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3. SYSTEM CALLS — 5 MARKS

Definition

A **system call** is a mechanism through which a user program requests a service from the operating system.

Types of System Calls

1. Process Control
Used to create, execute, and terminate processes.
Examples: fork(), exec(), exit().

2. File Management
Used to create, open, read, write, and close files.
Examples: open(), read(), write(), close().

3. Device Management
Used to request, use, and release I/O devices.

4. Information Maintenance
Used to obtain system information such as process ID and system time.

5. Communication
Used for communication between processes.
Examples: pipes and shared memory.

Diagram

User Program

↓

System Call

↓

Operating System

↓

Hardware

Conclusion

System calls provide an **interface between user programs and the operating system**.

---

4. PROCESSES — 5 MARKS

Definition

A **process is a program in execution**.

For example, when a program is loaded into memory and starts running, it becomes a process.

Components of a Process

A process contains:

1. Program code
2. Data
3. Stack
4. Heap
5. Program Counter
6. CPU registers

Process Control Block (PCB)

The OS maintains information about each process in a **PCB**.

PCB contains:

- Process ID
- Process state
- Program counter
- CPU registers
- CPU scheduling information
- Memory information
- I/O information

Diagram

Program

↓

Loaded into Memory

↓

Execution

↓

Process

Conclusion

A process is the **basic unit of execution** in an operating system.

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5. PROCESS STATES — 5 MARKS

A process passes through different states during its lifetime.

Five Process States

1. New
The process is being created.

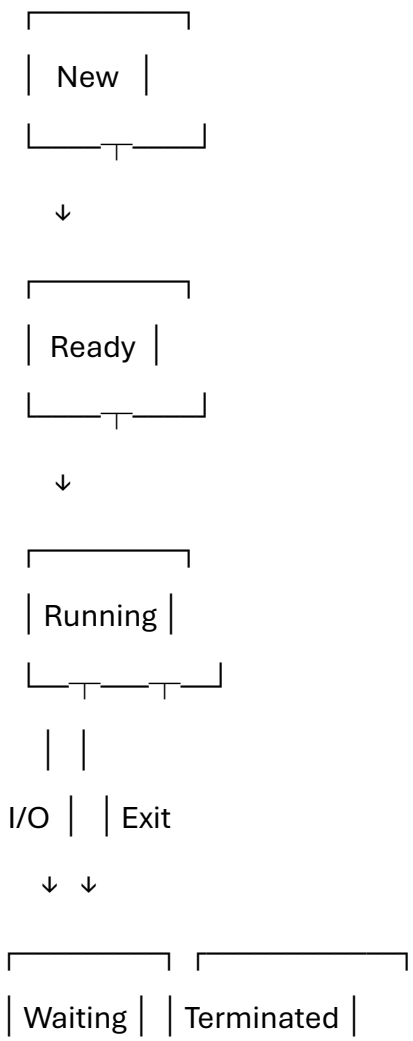
2. Ready
The process is waiting for the CPU.

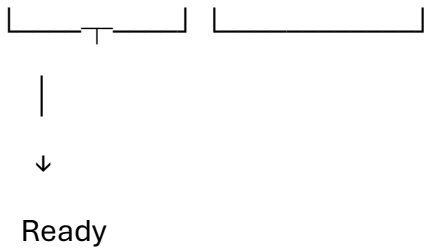
3. Running
The process is currently executing.

4. Waiting/Blocked
The process is waiting for an I/O operation or event.

5. Terminated
The process has completed its execution.

Diagram





Ready

Conclusion

Process states help the OS **manage and schedule processes efficiently**.

---

6. THREADS — 5 MARKS

Definition

A **thread is the smallest unit of CPU execution within a process**.

A process can contain one or more threads.

Types of Threads

1. User-Level Threads
These threads are managed by a user-level thread library.

2. Kernel-Level Threads
These threads are managed directly by the operating system kernel.

Advantages of Threads

1. Faster execution
2. Better CPU utilization
3. Less resource consumption
4. Easy communication between threads
5. Supports multitasking

Process vs Thread

Process	Thread
Heavyweight	Lightweight
Has separate memory	Shares process memory
Creation is slower	Creation is faster

Process	Thread
Communication is costly	Communication is easier

Conclusion

Threads improve performance by allowing **multiple activities to execute within the same process**.

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7. CPU SCHEDULING — 5 MARKS

Definition

CPU scheduling is the process of selecting a process from the ready queue and allocating the CPU to it.

CPU Scheduling Algorithms

1. FCFS (First Come First Serve)
The process that arrives first gets the CPU first.

2. SJF (Shortest Job First)
The process having the shortest burst time is executed first.

3. SRTF (Shortest Remaining Time First)
The process with the shortest remaining execution time is selected.

4. Priority Scheduling
The process with the highest priority is executed first.

5. Round Robin
Each process gets the CPU for a fixed time called **time quantum**.

Important Formulas

$$TAT = CT - AT$$
$$WT = TAT - BT$$

Where:

- **AT** = Arrival Time
- **BT** = Burst Time
- **CT** = Completion Time
- **TAT** = Turnaround Time
- **WT** = Waiting Time

Conclusion

CPU scheduling improves **CPU utilization, throughput, and response time**.

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8. OS AS A RESOURCE MANAGER — 5 MARKS

Definition

An operating system acts as a **resource manager** because it manages and allocates all computer resources.

Resources Managed by OS

1. CPU Management
The OS allocates CPU time to processes.

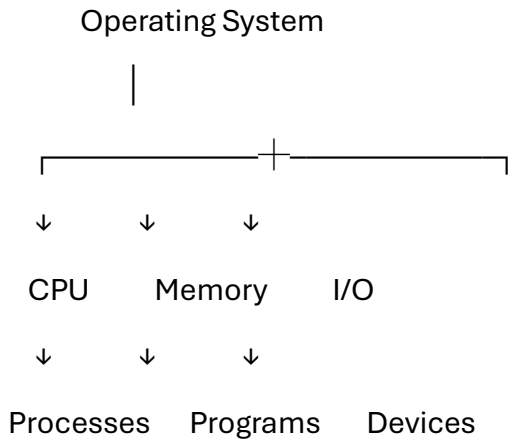
2. Memory Management
The OS allocates and deallocates memory.

3. File Management
The OS manages creation, deletion, reading, and writing of files.

4. I/O Device Management
The OS manages devices such as keyboard, printer, disk, and monitor.

5. Storage Management
The OS manages secondary storage such as HDD and SSD.

Diagram



Conclusion

The OS manages resources **efficiently, fairly, and safely** among different programs and users.

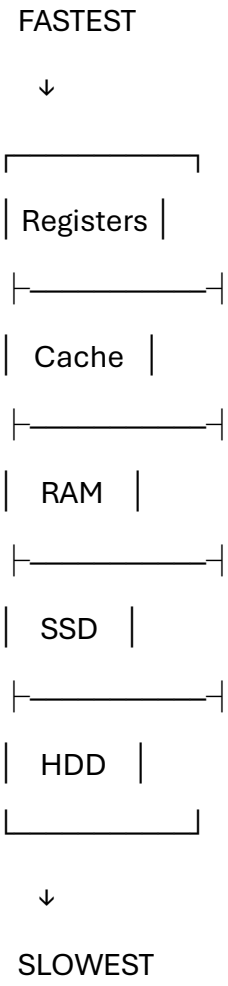
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9. MEMORY HIERARCHY — 5 MARKS

Definition

Memory hierarchy is the arrangement of different types of memory according to their **speed, cost, and capacity**.

Diagram



Characteristics

- **Registers** → fastest and smallest
- **Cache** → very fast memory
- **RAM** → main memory
- **SSD/HDD** → secondary storage
- Speed decreases as we move downward.
- Capacity generally increases as we move downward.
- Cost per bit generally decreases as we move downward.

Conclusion

Memory hierarchy provides a balance between **speed, storage capacity, and cost**.

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10. PROCESS SCHEDULING NUMERICALS — 5 MARKS

Scheduling numericals are **very important for the exam**.

Important Algorithms

1. FCFS
2. SJF
3. SRTF
4. Priority Scheduling
5. Round Robin

Important Terms

Completion Time (CT):
Time at which a process completes.

Turnaround Time (TAT):

$$TAT = CT - AT$$

Waiting Time (WT):

$$WT = TAT - BT$$

Example: FCFS

Given:

Process	AT	BT
P1	0	5
P2	1	3
P3	2	2

Gantt Chart

0 5 8 10

| P1 | P2 | P3 |

Completion Time

- P1 = 5
- P2 = 8
- P3 = 10

Turnaround Time

P1:

$$5 - 0 = 5$$

P2:

$$8 - 1 = 7$$

P3:

$$10 - 2 = 8$$

Waiting Time

P1:

$$5 - 5 = 0$$

P2:

$$7 - 3 = 4$$

P3:

$$8 - 2 = 6$$

Average Waiting Time

$$AWT = \frac{0 + 4 + 6}{3}$$

$$\boxed{AWT = 3.33}$$

Average Turnaround Time

$$ATAT = \frac{5 + 7 + 8}{3}$$

$$\boxed{ATAT = 6.67}$$

Exam order:
Given data → Gantt chart → CT → TAT → WT → Average.

⭐ Most Important 5-Mark Questions

Before your exam, make sure you can write these without looking:

1. **Explain types of Operating Systems.**
2. **Explain different OS structures.**
3. **Explain system calls and their types.**
4. **Define process and explain PCB.**
5. **Explain process states with diagram.**
6. **Explain threads and their types.**
7. **Explain CPU scheduling algorithms.**
8. **Explain OS as a resource manager.**
9. **Explain memory hierarchy with diagram.**
10. **Solve CPU scheduling numericals using FCFS, SJF, Priority and Round Robin.**